

# Full Stack Development with MERN

## Project Documentation format

### 1. Introduction

- **Project Title:** Implement Client Script & UI Policy (Incident)
- **Team Members:** To be added by the team (names and roles were not provided in the project source).

### 2. Project Overview

- **Purpose:** Enforce data integrity on ServiceNow Incident records by using UI Policies and Client Scripts, so incidents are complete, valid and consistent before they are submitted. This improves triage, routing, reporting accuracy and SLA compliance.
- **Features:**
  - UI Policy *High Impact Control* (Impact is 1 - High) with *Reverse if false* enabled.
  - UI Policy Action: Assignment group becomes mandatory.
  - UI Policy Action: Urgency becomes read-only.
  - onChange Client Script: Urgency is set to 1 - High when Impact changes to High, with an info message.
  - onSubmit Client Script: save is blocked and an error box is shown when Assigned to is empty for a High impact incident.
  - onCellEdit Client Script: State cannot be changed from the list view.

### 3. Architecture

- **Frontend:** The ServiceNow web UI (Incident form and list view, Desktop UI type). The UI Policy and the three Client Scripts run in the browser using JavaScript and the g\_form API. React is not used in this project.
- **Backend:** The ServiceNow Now Platform hosts and serves the Incident application. No custom server-side code was written, so Node.js and Express.js are not used; the controls are purely client-side.
- **Database:** ServiceNow's built-in relational database. The project works with the Incident table [incident], using the fields Number, Caller, Category, Impact, Urgency, Priority, State, Assignment group, Assigned to and Short description. MongoDB is not used.

### 4. Setup Instructions

- **Prerequisites:** A ServiceNow instance (this project used dev354432.service-now.com) with administrative or configuration access, the Incident Management module, and a web browser. Node.js and MongoDB are not required.
- **Installation:** Configuration steps on the instance:
  - Log in with administrative access and open System UI > UI Policies > New.
  - Create the UI Policy: Name = High Impact Control, Table = Incident, Active = true; condition Impact is 1 - High; Reverse if false = true.

- o Add the UI Policy Actions: Assignment group (Mandatory = True) and Urgency (Read only = True, Visible = Leave alone).
- o Open System UI > Client Scripts > New and create the onChange script (Impact), the onSubmit script and the onCellEdit script (State) on the Incident table, all Active.
- o Paste the scripts from the appendix of the project report and click Submit for each.

## 5. Folder Structure

- **Client:** ServiceNow has no project folders; the client-side artifacts are records on the instance:  

```

System UI > UI Policies
+-- High Impact Control                (Incident)
    +-- Action: assignment_group        (Mandatory = True)
    +-- Action: urgency                (Read only = True)
System UI > Client Scripts
+-- Auto set urgency for high impact    (onChange, Impact)
+-- Prevent save if Assigned To missing (onSubmit)
+-- Prevent state change via list edit  (onCellEdit, State)

```
- **Server:** No custom server-side files were created. Storage and processing are handled by the platform and the Incident table.

## 6. Running the Application

- There are no start commands; ServiceNow runs as a hosted service:
  - o **Frontend:** Open the instance URL in a browser, then go to Incident > Create New (form) or Incident > All (list).
  - o **Backend:** Managed by the ServiceNow platform; nothing needs to be started locally.

## 7. API Documentation

- No custom endpoints were created or exposed by this project.
- The platform's standard Table API for the Incident table is available (illustrative example below). Client-side UI Policies and Client Scripts do not run for REST/API calls, so these rules are not enforced there.

| Method | Endpoint                         | Parameters                                      | Example response                                       |
|--------|----------------------------------|-------------------------------------------------|--------------------------------------------------------|
| POST   | /api/now/table/incident          | JSON body, e.g.<br>short_description,<br>impact | { "result": { "number":<br>"INC...", "impact": "1" } } |
| GET    | /api/now/table/incident          | sysparm_query,<br>sysparm_limit                 | { "result": [ { "number":<br>"INC..." } ] }            |
| PATCH  | /api/now/table/incident/{sys_id} | JSON body with<br>changed fields                | { "result": { "sys_id": "..." } }                      |

## 8. Authentication

- Access uses ServiceNow's standard login; the project was configured with an administrative account, and incidents were tested through the normal UI session.

- Authorization is role-based on the platform (administrative access to create UI Policies and Client Scripts). The project adds no custom tokens, sessions or authentication logic.

## 9. User Interface

- Configuration screens of the UI Policy, UI Policy Actions and Client Scripts:

The screenshot shows the 'UI Policy - High Impact Control' configuration page. The left sidebar contains a navigation menu with categories like 'System Mobile', 'Mobile UI', 'Service Catalog', 'Catalog Administration', 'Catalog UI Policies', 'System Diagnostics', 'Session Debug', 'Debug UI Policies', 'Disable UI Policies Debug', 'System UI', 'UI Policies', 'MID Server', 'MID Security Policy', 'System Web Services', and 'SOAP Security Policies'. The main content area is titled 'UI Policy - High Impact Control View: Advanced'. It includes a 'Table' dropdown set to 'Incident [Incident]', an 'Application' dropdown set to 'Global', and an 'Active' checkbox checked. A 'Short description' field contains 'High Impact Control' and an 'Order' field contains '100'. The 'When to Apply' section has a 'Script' tab and a 'Conditions' section with a rule: 'Impact' is '1 - High'. Below this, there are checkboxes for 'Global' (checked), 'Reverse if false' (checked), and 'On load' (checked). A note states: 'If selected, the UI Policy applies to all form views; otherwise the UI Policy is view-specific'. Another note says: 'Apply the UI policy actions when the form is loaded and when the user changes values on the form'. A final note mentions: 'Tables that extend the specified Table inherit this UI Policy'.

Figure 1: UI Policy High Impact Control (Incident, Impact is 1 - High, Reverse if false).

The screenshot shows the 'UI Policy Action - assignment\_group' configuration page. The left sidebar is the same as in Figure 1. The main content area is titled 'UI Policy Action - assignment\_group'. It includes a 'UI policy' dropdown set to 'High Impact Control', a 'Table' dropdown set to 'Incident [Incident]', and a '\* Field name' dropdown set to 'Assignment group'. On the right, there are three dropdowns: 'Mandatory' set to 'True', 'Visible' set to 'Leave alone', and 'Read only' set to 'False'. There is also a 'Clear the field value' checkbox. At the bottom, there are 'Update' and 'Delete' buttons.

Figure 2: UI Policy Action - Assignment group (Mandatory = True).

The screenshot shows the 'UI Policy Action - urgency' configuration page. The left sidebar is the same as in Figure 1. The main content area is titled 'UI Policy Action - urgency'. It includes a 'UI policy' dropdown set to 'High Impact Control', a 'Table' dropdown set to 'Incident [Incident]', and a '\* Field name' dropdown set to 'Urgency'. On the right, there are three dropdowns: 'Mandatory' set to 'Leave alone', 'Visible' set to 'Leave alone', and 'Read only' set to 'True'. There is also a 'Clear the field value' checkbox. At the bottom, there are 'Update' and 'Delete' buttons.

Figure 3: UI Policy Action - Urgency (Read only = True).

**Client Script - Auto set urgency for high impact**

Name: Auto set urgency for high impact

Table: Incident [Incident]

UI Type: Desktop

Type: onChange

Field name: Impact

Application: Global

Active: ☒

Inherited: ☐

Global: ☒

Script:

```

1 function onChange(control, oldValue, newValue, isLoading) {
2   if (isLoading) newValue = '';
3   return;
4 }
5
6 if (newValue == '1') { // High Impact
7   g_form.setFieldValue('urgency', '1');
8   g_form.showErrorMessage(
9     'Impact',
10    'High Impact Incidents must have High urgency and an assignment group.',
11    'info'
12  );
13 }
14 }
15

```

Isolate script: ☒

Figure 4: onChange Client Script - Auto set urgency for high impact.

**Client Script - Prevent save if assigned to missing**

Name: Prevent save if assigned to missing

Table: Incident [Incident]

UI Type: Desktop

Type: onSubmit

Application: Global

Active: ☒

Inherited: ☐

Global: ☒

Script:

```

1 function onSubmit() {
2   if (g_form.getValue('impact') == '1' &&
3       g_form.getValue('assigned_to') == '') {
4     g_form.showErrorBox(
5       'assigned_to',
6       'assigned_to is mandatory for High impact Incidents.'
7     );
8     return false;
9   }
10  return true;
11 }
12
13
14

```

Isolate script: ☒

Figure 5: onSubmit Client Script - Prevent save if Assigned To missing.

**Client Script - Prevent state change via list edit**

Name: Prevent state change via list edit

Table: Incident [Incident]

UI Type: Desktop

Type: onCellEdit

Field name: State

Application: Global

Active: ☒

Inherited: ☐

Global: ☒

Script:

```

1 function onCellEdit(sysId, table, oldValues, newValue, callback) {
2
3   alert('State cannot be changed from the list. Please open the Incident.');


4



5   // Cancel the list edit



6   callback(false, oldValues);



7 }



8



9



Isolate script: ☒


```

Figure 6: onCellEdit Client Script - Prevent state change via list edit.

## 10. Testing

- **Strategy:** manual functional (user acceptance) testing on the Incident form and list, with positive and negative scenarios, carried out on 14 January 2026.
- **Tools:** ServiceNow Incident form and list view in a web browser.
- **Result:** 10 test cases, all passed (0 failed, 0 not tested); one low-severity observation closed as by design.

| ID     | Test scenario                                             | Result |
|--------|-----------------------------------------------------------|--------|
| TC-001 | Assignment group mandatory when Impact = High (UI Policy) | Pass   |
| TC-002 | Urgency read-only when Impact = High (UI Policy Action)   | Pass   |
| TC-003 | Urgency auto-set for High impact (onChange)               | Pass   |
| TC-004 | Negative: save without Assigned to (onSubmit)             | Pass   |
| TC-005 | Positive: save with Assigned to filled (onSubmit)         | Pass   |
| TC-006 | State edit blocked in list view (onCellEdit)              | Pass   |
| TC-007 | Reverse if false: Assigned to not mandatory               | Pass   |
| TC-008 | Reverse if false: Urgency becomes editable                | Pass   |
| TC-009 | Record saves after Impact changed to Medium               | Pass   |
| TC-010 | State change from the Incident form                       | Pass   |

## 11. Screenshots or Demo

- Test evidence from the instance (no separate demo link was provided):

The screenshot shows the 'Incident - Create INC0010012' form in ServiceNow. The 'Impact' dropdown is set to '1 - High'. The 'Assigned to' field is empty, and a red error message is displayed: 'Assigned To is mandatory for High impact incidents.' The 'Submit' button is highlighted with a red box.

Figure 7: Save blocked; error on Assigned to for High impact.

The screenshot shows the same 'Incident - Create INC0010012' form, but now the 'Assigned to' field is filled with 'Beth Anglin'. The error message is no longer present, and the 'Submit' button is still highlighted with a red box.

Figure 8: Incident saved after Assigned to is filled.

ServiceNow Incident form for INC0010012. The form includes fields for Number, Caller, Category, Subcategory, Service, Configuration Item, Short description, and Description. On the right, there are dropdowns for Channel, State, Impact, Urgency, Priority, Assignment group, and Assigned to. Red boxes highlight the Impact (2 - Medium), Urgency (1 - High), and Assigned to fields.

Figure 9: Reverse condition; Impact changed to Medium.

ServiceNow Incidents list view. A modal dialog is displayed with the message: "dev354432.service-now.com says State cannot be changed from the list. Please open the Incident." The list shows various incidents with columns for Number, Opened, Category, Assignment group, Assigned to, and Updated by.

| Number     | Opened              | Category       | Assignment group | Assigned to | Updated by |
|------------|---------------------|----------------|------------------|-------------|------------|
| INC0010012 | 2026-01-14 02:15:14 | Inquiry / Help | Software         | admin       | admin      |
| INC0009009 | 2018-08-30 01:06:16 | Inquiry / Help | Software         | admin       | admin      |
| INC0009005 | 2018-08-31 21:35:21 | Inquiry / Help | Software         | admin       | admin      |
| INC0009001 | 2018-09-11 20:56:26 | Inquiry / Help | Software         | admin       | admin      |
| INC0008112 | 2019-07-29 11:48:43 | Inquiry / Help | Software         | admin       | admin      |
| INC0008111 | 2019-07-22 14:04:57 | Inquiry / Help | Software         | admin       | admin      |
| INC0008001 | 2021-01-15 13:04:14 | Inquiry / Help | Software         | admin       | admin      |
| INC0007002 | 2018-10-16 22:47:51 | Inquiry / Help | Software         | admin       | admin      |
| INC0007001 | 2018-10-16 22:47:10 | Inquiry / Help | Software         | admin       | admin      |
| INC0001990 | 2020-06-07 09:02:25 | Inquiry / Help | Software         | admin       | admin      |
| INC0001990 | 2020-06-07 09:02:25 | Inquiry / Help | Software         | admin       | admin      |
| INC0001990 | 2020-06-07 09:02:25 | Inquiry / Help | Software         | admin       | admin      |

Figure 10: Alert when State is edited from the list.

ServiceNow Incident form for INC0008111. The form includes fields for Number, Caller, Category, Subcategory, Service, Configuration Item, Short description, and Description. On the right, there are dropdowns for Channel, State, Impact, Urgency, Priority, Assignment group, and Assigned to. Red boxes highlight the Number, State, and Impact fields. A context menu is open over the State dropdown, showing options like "Add to Visual Task Board", "Copy Incident", "Create Incident Task", "Create Outage", "Create Problem", "Create Request", "Create Child Incident", "Create Normal Change", "Create Standard Change", "Create Emergency Change", "Refresh Impacted Services", "Metrics Timeline", "Follow on Live Feed", "Analyze Access", "Configure", "Export", "View", "Create Favorite", "Copy URL", "Copy sys\_id", "Show XML", "History", and "Reload form".

Figure 11: State updated on the Incident form.

| Number     | Opened              | Short description                                               | Caller                 | Priority     | State       | Category       | Assignment group | Assigned to | Updated             | Updated by |
|------------|---------------------|-----------------------------------------------------------------|------------------------|--------------|-------------|----------------|------------------|-------------|---------------------|------------|
| INC0010012 | 2020-01-14 02:15:14 | testing UI policy                                               | Adela Cervantiz        | 2 - High     | New         | Inquiry / Help | Software         | (empty)     | 2020-01-14 03:19:11 | admin      |
| INC0009009 | 2018-08-30 01:06:16 | Unable to access the shared folder.                             | David Miller           | 4 - Low      | New         | Inquiry / Help | (empty)          | (empty)     | 2018-12-12 23:30:24 | admin      |
| INC0009005 | 2018-08-31 21:35:21 | Email server is down.                                           | David Miller           | 1 - Critical | New         | Software       | (empty)          | (empty)     | 2018-12-12 23:18:55 | admin      |
| INC0009001 | 2018-09-11 20:56:26 | Unable to post content on a Wiki page                           | David Miller           | 3 - Moderate | New         | Inquiry / Help | (empty)          | (empty)     | 2018-12-12 23:32:42 | admin      |
| INC0008112 | 2019-07-29 11:48:43 | Assessment : ATF Assessor                                       | survey user            | 5 - Planning | New         | Inquiry / Help | (empty)          | (empty)     | 2019-07-29 11:49:28 | admin      |
| INC0008111 | 2019-07-22 14:04:57 | ATF - Test1                                                     | System Administrator   | 5 - Planning | In Progress | Inquiry / Help | (empty)          | (empty)     | 2020-01-14 03:31:59 | admin      |
| INC0008001 | 2021-01-15 13:04:14 | ATF:TEST2                                                       | survey user            | 5 - Planning | New         | Inquiry / Help | (empty)          | (empty)     | 2021-01-21 15:31:42 | admin      |
| INC0007002 | 2018-10-16 22:47:51 | Need access to the common drive.                                | David Miller           | 4 - Low      | New         | Inquiry / Help | (empty)          | (empty)     | 2018-12-12 23:28:49 | admin      |
| INC0007001 | 2018-10-16 22:47:10 | Employee payroll application server is down.                    | David Miller           | 1 - Critical | New         | Hardware       | Openspace        | (empty)     | 2025-12-03 18:11:31 | system     |
| INC0001990 | 2020-06-07 09:02:25 | Unable to access the personal details section in payroll portal | Problem CoordinatorATF | 5 - Planning | On Hold     | Inquiry / Help | (empty)          | (empty)     | 2020-06-07 09:04:34 | admin      |
| INC0000000 | 2016-08-10          | Unable to access team                                           | Philipp R. ...         | 3 - ...      | ...         | ...            | ...              | ...         | 2016-08-10          | ...        |

Figure 12: Incident list after the form update.

## 12. Known Issues

- When Impact is lowered from High to Medium, Urgency keeps the value 1 - High (the onChange script only acts on High); it becomes editable and must be changed manually.
- The controls are client-side only and do not apply to API calls, imports or other server-side updates.
- The onCellEdit script blocks State edits in the list for all users, including legitimate ones.
- The onChange script shown in the instance screenshot (field message) differs slightly from the script in the instructions (info message); keep one version.

## 13. Future Enhancements

- Add a server-side Business Rule or Data Policy so the rules also cover API, import and integration updates.
- Reset or review Urgency automatically when Impact is lowered.
- Build a dashboard to monitor incidents with missing or inconsistent data.
- Replace hard-coded values with system properties and add role-based exceptions for list edits.